

Introduction:

The current state of Austin's water quality is a cause for concern, despite the city's efforts to manage and protect its water resources ([Austin Water](#)). This contamination includes fecal bacteria, often caused by Sewage spills, sewage overflows, and contaminated stormwater ([nih](#)).

Fecal contamination in water bodies can increase risk of illness in humans through contact during recreational activities such as swimming and wading ([Watershed Protection](#)), and heavy metals can cause harm to various organs, even at low exposure levels ([nih](#)). Studies have shown that socioeconomic disparities can be related to exposure to contaminated water ([nih](#)). We aim to show the spatial correlation between bacterial water contamination and median income level.

Methods:

- Used the Water Quality Sampling data set from the city of Austin Water Resources Monitoring database
- Filtered for surface water and groundwater
- Filtered for parameters: arsenic, cadmium, chromium, copper, iron, lead, lithium, nickel, zinc, E. coli bacteria, and total coliform bacteria
- Used 2015 samples as training data, 2016, as testing data
- retrieved zip code and median household income data from US Census Data
- classified zip codes as low or high income based on whether the median incomes were below or above the Texas annual median household income for 2015 and 2016 respectively, and hot coded low income as 1 and high income as 0

Research Question:

Given the water sampling frequency and medium result of E. coli sampling from watersheds in a ZIP code in Austin TX, can we predict whether it belongs to a high or low income area?

Model iteration #1: Multiple Logistic Regression

While our training data set includes 11 different measurements of median values of contaminants, the majority of watersheds in our data set were not tested for all 11 contaminants. As multiple logistic regression does not work well with so many N/A values, we could only use total frequency of sampling and median e coli values in order to estimate whether the Zipcode the water sample came from was a low income or high income area.

```
Call:
glm(formula = lowincome ~ total_frequency + median_ecoli, family = binomial,
    data = zip_training)
```

Coefficients:

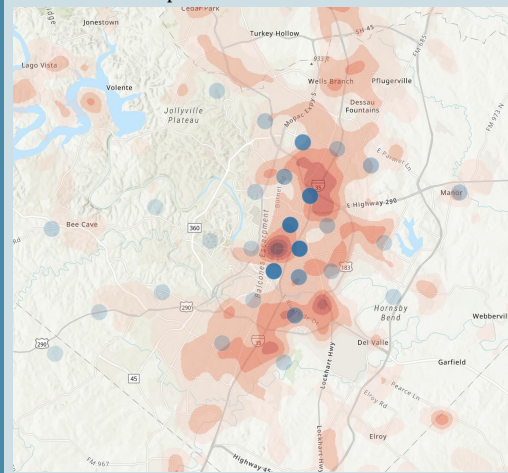
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.4225751	0.7150977	0.591	0.555
total_frequency	-0.0237933	0.0183852	-1.294	0.196
median_ecoli	-0.0001198	0.0007513	-0.159	0.873

$$p(\text{lowincome}) = \hat{\beta}_0 + \hat{\beta}_1 \text{frequency} + \hat{\beta}_2 \text{ecoli}$$

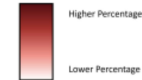
The model with the two predictors estimating income level demonstrates that there is a negative relationship between frequency of testing and whether a watershed is from a low income area, and that there is also a negative relationship between median amount of e coli in a water sample and the water sample's zip code being low income.

While we used a Multiple Logistic Regression model, due to the volume of N/A values for the majority of the water contaminants we intended to include as predictors, we will iterate upon this model by using indicator terms that allow us to dismiss a contaminant if it has an N/A value for a zip code, but include it if there is a measurement that can aid in the accuracy of our prediction.

Median E. Coli per ZIP vs Low Household Income



Austin Annual HHI Below \$50,000



Performance of our Statistical Learning
Method: 54% (for now) ... better than random guessing!

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	13	5
1	12	7

Accuracy : 0.5405
95% CI : (0.3692, 0.7051)
No Information Rate : 0.6757
P-Value [Acc > NIR] : 0.9707

